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Impacts of Predicted Liquid Fraction and Multiple Ice-Phase Categories on the Simulation of Hail in the Predicted Particle Properties (P3) Microphysics Scheme

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Key Points:

- Predicting the liquid fraction improves the parameterization of melting and results in reduced hail sizes
- The use of multiple free ice-phase categories reduces the dilution of bulk physical properties of ice and increases hail sizes
- With recent updates, the Predicted Particle Properties bulk microphysics scheme is well-suited for modeling hailstorms

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Abstract Since its inception in 2015, the Predicted Particle Properties (P3) bulk microphysics scheme has undergone several major developments. Ice is now represented by a user-specified number of freely-evolving (non-prescribed) categories; the liquid fraction of particles is predicted, thereby allowing for mixed-phase particles and improved process rates; and the scheme is triple-moment, which allows the size spectral width to vary independently. As such, P3 is now capable of representing key properties and microphysical processes that are important for hail. In this study, the impacts of some new capabilities of P3 on the simulation of hail amounts and sizes are examined in the context of idealized, high-resolution (200-m isotropic grid spacing) hailstorm simulations using a cloud-resolving model. Sensitivity tests are conducted to examine the impacts of (a) the predicted liquid fraction, and (b) the number of generic ice-phase categories (varied between one and four). Predicted liquid fraction leads to a more realistic treatment of melting and shedding, which decreases the mean ice (hail) sizes during melting compared to the original P3 scheme. In contrast, with an increasing number of ice-phase categories, the problem of property dilution is mitigated, ultimately resulting in greater quantities of hail and larger sizes reaching the surface. It is argued that the latest version of the P3 scheme is now capable of realistically representing the major microphysical processes involved in the initiation, growth, and decay of hail.

Plain Language Summary The Predicted Particle Properties (P3) scheme is a piece of computer code, used in atmospheric numerical models, that calculates the bulk effects of cloud processes that ultimately lead to precipitation. Since its inception in 2015, the P3 scheme has undergone several major developments. Ice-phase particles are now represented by a user-specified number of freely evolving generic categories; P3 now predicts the liquid fraction which allows for mixed-phase particles and more accurate calculations of physical processes; and there is flexibility in representing the distribution of ice particle sizes. With these features, P3 is now capable of simulating the physical processes related to the growth and decay of hail. In this study, the impacts of the new features on the simulation of hail were examined using a fine-scale, detailed numerical model capable of simulating individual storms. The predicted liquid fraction was shown to improve the simulation of hail melting and resulted in smaller hail at the surface. In contrast, increasing the number of ice-phase categories ultimately allows for the simulation of larger hail sizes.

1. Introduction

A major component of an atmospheric model is the bulk microphysics scheme (BMS), which parameterizes the effects of grid-scale clouds and precipitation. Despite the existence of more detailed approaches, bin-resolving and Lagrangian super-particles schemes, BMSs continue to play important roles in research and are ubiquitous in operational numerical weather and climate prediction models. This is expected to continue into the foreseeable future. Since ice was introduced into single-moment schemes decades ago (e.g., Rutledge & Hobbs, 1983), the treatment of ice-phase microphysics has become increasingly sophisticated. Over the years, more categories were added to expand the range of properties of frozen particles represented in models, in particular terminal fall speeds (e.g., Lin et al., 1983). Detailed two-moment schemes later become popular (e.g., Ferrier, 1994; Meyers et al., 1997; Morrison et al., 2005, 2009; Murakami, 1990; Reisner et al., 1998; Seifert & Beheng, 2006; Ziegler, 1985). Three-moment schemes then followed (Chen & Tsai, 2016; Dawson et al., 2014; Loftus et al., 2014; Milbrandt & Yau, 2005a, 2005b; Yang & Yau, 2008). More recently, the prognostic volume mixing ratio for rimed ice categories (graupel and hail) was included in some schemes, which allows for the prediction of the bulk particle density (Jensen et al., 2023; Mansell et al., 2010; Milbrandt & Morrison, 2013; Park et al., 2024). Tsai and

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Chen (2020) developed a BMS which, in addition to bulk density, predicts the shape evolution of the ice-phase categories. Thompson and Eidhammer (2014) included prognostic aerosols, which ultimately is important in determining droplet sizes; this in turn impacts the rime density in the most recent version of this scheme, which now includes prognostic graupel density. The prediction of the liquid fraction for hail was included in Loftus et al. (2014), via prognostic heat content, and for partly-frozen hydrometeors in general in Cholette et al. (2019), via prognostic liquid of mixed-phase particles. With all of these predictive aspects, BMSs now have the capability, in principle, of simulating the key microphysical processes involved in the production, growth, and decay of hail.

Despite many advances, most BMS are still constructed around the original approach of using pre-defined ice-phase categories (e.g., "snow," "graupel," etc.) usually with fixed parameters to define or diagnose the values of bulk physical properties. This traditional approach has some inherent and insurmountable weaknesses. First, the properties of natural ice particles vary considerably. This is in contrast with liquid drops and droplets, whose density is constant and with spherical shape (except for large drops). Perhaps most importantly, the traditional approach requires the parameterized process of conversion between categories; this process is unphysical and is a necessary artifact of the construction of the scheme. It is therefore unconstrained by observations or theory and its closure assumptions introduce model sensitivity (Morrison et al., 2020). Note that these problems are equally inherent to most bin microphysics schemes, which is illustrated in Xue et al. (2017).

An alternative to traditional, pre-defined ice category-based schemes is the property-based approach, which uses one or more ice-phase categories that do not correspond a priori to a particular ice type. That is, categories are not assigned a type with corresponding fixed parameters, but rather each category includes a set of prognostic variables from which physical properties can be derived and vary continuously in time and space. The first property-based microphysics scheme can probably be credited to Hashino and Tripoli (2007), though this was a bin scheme. The Predicted Particle Properties (P3) scheme was introduced by Morrison and Milbrandt (2015), who proposed the concept of a generic, freely-evolving, bulk ice-phase category. Harrington et al. (2013) and Jensen et al. (2017) also constructed a property-based bulk scheme, now called Ice-Spheroids Habit Model with Aspect-Ratio Evolution (ISHMAEL), with a unique capacity to predict the evolution of the aspect ratios of ice crystals. To our knowledge, these are the two only property-based BMSs that have been developed that completely eschew traditional ice categories.

Although the P3 scheme was originally constructed as a proof of concept, it initially performed well against detailed, mature, well-calibrated, traditional BMSs. This was true for both historic case studies (Morrison et al., 2015) and in a real-time experimental numerical weather prediction (NWP) context (see Milbrandt et al., 2021). It was (and is) also computationally efficient due in part to its extensive use of pre-computed lookup tables during run time (Morrison et al., 2015). It has since been interfaced with several mesoscale and large-eddy simulation models; it has been used operationally since 2018 in the 2.5-km deterministic NWP system of Environment and Climate Change Canada (ECCC) (Milbrandt et al., 2016, 2018) and it is available in the Weather Research and Forecasting model (WRF; Skamarock et al., 2008), the Cloud Model 1 (CM1) cloud-resolving model (Bryan & Fritsch, 2002), the Energy Exascale Earth System Model (E3SM) climate model (Wang et al., 2021), and other models. Since its inception, P3 has undergone several major developments. This includes the generalization to multiple ice-phase categories (Milbrandt & Morrison, 2016), prognostic liquid fraction of particles (Cholette et al., 2019), a triple-moment representation of ice (Milbrandt et al., 2021), and a diagnostic subgrid-scale cloud fraction (Jouan et al., 2020). A triple-moment rain version was also developed (Paukert et al., 2019) though this was not incorporated into the main P3 code stream.

Although the P3 scheme is now quite mature and its representation of ice is highly versatile, some aspects have not yet been examined with regards to their impacts on the simulation of hail. In a study using idealized supercell simulations, Johnson et al. (2019) compared the performance of P3 to that of two detailed traditional BMSs and found P3 to perform poorly in comparison for the simulation of dual-polarization radar signatures for hail. These deficiencies were alleviated by the introduction of triple-moment ice (and also improved for a modified double-moment ice configuration) in Milbrandt et al. (2021), which relaxed the restriction of mean ice sizes by varying the particle size spectral width and controlling gravitational size sorting. The impacts of other potentially important recent innovations to P3 have not yet been explored.

The goal of this study, therefore, is to examine the following question: What are the impacts on the simulation of hail from (a) the predicted liquid fraction and (b) multiple generic ice categories? This is examined using idealized

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Table 1
List of Simulation Configurations

Run name	*nCat*	Liq-Frac	SIP	Treatment of $(dN_i/dt)_{\mathrm{MELT}}$
BASE	1	Off	Off	Proportional to Q_i/N_i
LF	1	On	Off	See text
BASE-MOD	1	Off	Off	$(dN_i/dt)_{\mathrm{MELT}} = 0$ (similar to LF)
LF-MOD1	1	On	Off	Identical to BASE
LF-MOD2	1	On	On	Identical to LF
LF-2CAT	2	On	On	Identical to LF
LF-3CAT	3	On	On	Identical to LF
LF-4CAT	4	On	On	Identical to LF

Note. The baseline configuration uses one ice category, has predicted liquid fraction off, and SIP off. "LF" denotes liquid fraction on. "MOD(*x*)" denotes modified; see main text for details. For multi-ice-category runs, (*x*) in "LF-(*x*) CAT" indicates the number of ice categories. All simulations use triple-moment ice.

high-resolution (200-m grid spacing) cloud model simulations using the latest version (v5) of P3 for a deep convective hail-producing storm.

The remainder of the paper is organized as follows: Section 2 describes the experimental design, including the cloud model configuration and the idealized hailstorm case. Section 3 summarizes the control simulation. Section 4 presents the results of two sensitivity experiments, one looking at the effects of liquid fraction and the other on using different numbers of ice categories. Discussion is provided in Section 5 with concluding remarks in Section 6.

2. Experimental Design

2.1. Overview of the P3 Scheme

A complete description of the original single-category version of the P3 scheme is provided in Morrison and Milbrandt (2015) and an overview of the most recent version (at the time of writing), with details on how triple-moment ice and predicted liquid fraction were combined, is given in Cholette et al. (2023). The liquid-phase component of P3 is a standard two-category ("*cloud*" droplets and "*rain*"), two-moment (prognostic mass and number) formulation, with a gamma particle size distribution (PSD) for each category. Ice is represented with a user-specified number (*nCat*; all symbols are defined in Table A1) of generic ice-phase categories. The PSD of each ice category is also represented by a gamma function. For all hydrometeor categories, the PSD is of the form

$$N(D) = N_0 D^{\mu} e^{-\lambda D}, \tag{1}$$

where $N(D)$ is the number of particles between D and $D + dD$, D is the equivalent diameter, and N_0, λ, and μ are the intercept, slope, and shape parameters, respectively (see Table 1 for units). There are between four and six prognostic mixing ratio variables per ice category: the total mass ($Q_{i,\mathrm{tot}}$), rime mass ($Q_{i,\mathrm{rim}}$), total number ($N_{i,\mathrm{tot}}$), rime volume ($B_{i,\mathrm{rim}}$), and optionally the sixth moment (Z_i) and liquid mass accumulated on ice ($Q_{i,\mathrm{liq}}$). Since all of these quantities are mixing ratios, they are conserved and can be advected and diffused appropriately. Note, $Q_{i,\mathrm{tot}}$ is the total mass of a given ice (in general, mixed-phase) category, n, equal to the sum of $Q_{i,\mathrm{liq}}$, $Q_{i,\mathrm{rim}}$, and the deposition mass (which can be computed but is not actually a prognostic variable) for the category. The actual total mass of ice- or mixed-phase particles in a grid element is the sum of all $Q_{i,\mathrm{tot}}(n)$ for n from 1 to *nCat*.

From these prognostic variables, various bulk physical properties can be computed, including the rime fraction (F_{rim}), liquid fraction (F_{liq}), density (ρ_i), mass-weighted mean terminal fall speed (V_m) (and also the number-weighted and sixth moment-weighted fall speeds, for the sedimentation of $N_{i,\mathrm{tot}}$ and $Z_{i,\mathrm{tot}}$, respectively), and mass-weighted mean diameter (D_m). In this study, we also examine the maximum hail size, $D_{h,\mathrm{max}}$, which can be diagnosed from the tail of the PSD following the method described in Milbrandt et al. (2021).

An overview of the representation of ice (and mixed-phase particles) and the major growth/decay processes is depicted in Figure 1. Pristine crystals can be initiated from either primary nucleation, homogeneous freezing of cloud droplets, or secondary ice production (SIP). Currently the only SIP mechanism included is rime splintering, but in the near future other modes will be added, such as the fragmentation of freezing drops, which was shown to be important for tropical convection (see Qu et al., 2022). Ice can grow (or decay) from deposition (sublimation), undergo aggregation, both within a given ice category and between categories, and grow gradually from riming. With the predicted liquid fraction, liquid can remain on the particles during wet growth and possibly re-freeze before being shed as rain. During melting, the small ice sizes in the PSD melt instantly to rain while the remaining ice mass is retained as partially-melted mixed-phase particles. Shedding during melting and wet growth is computed explicitly with the prediction of liquid fraction. Mixed-phase particles can also refreeze if they fall or are transported into subfreezing air. As in the original scheme, rain can freeze to solid (maximum-density) ice via either immersion freezing or through collisions with existing ice. Thus, hail embryos can originate from either graupel or frozen drops, and since freezing has a drop-size dependence there is an indirect dependence on aerosols for the origin of hail embryos. (Currently P3 does not have prognostic aerosol concentrations, though this will be implemented in the near future.)

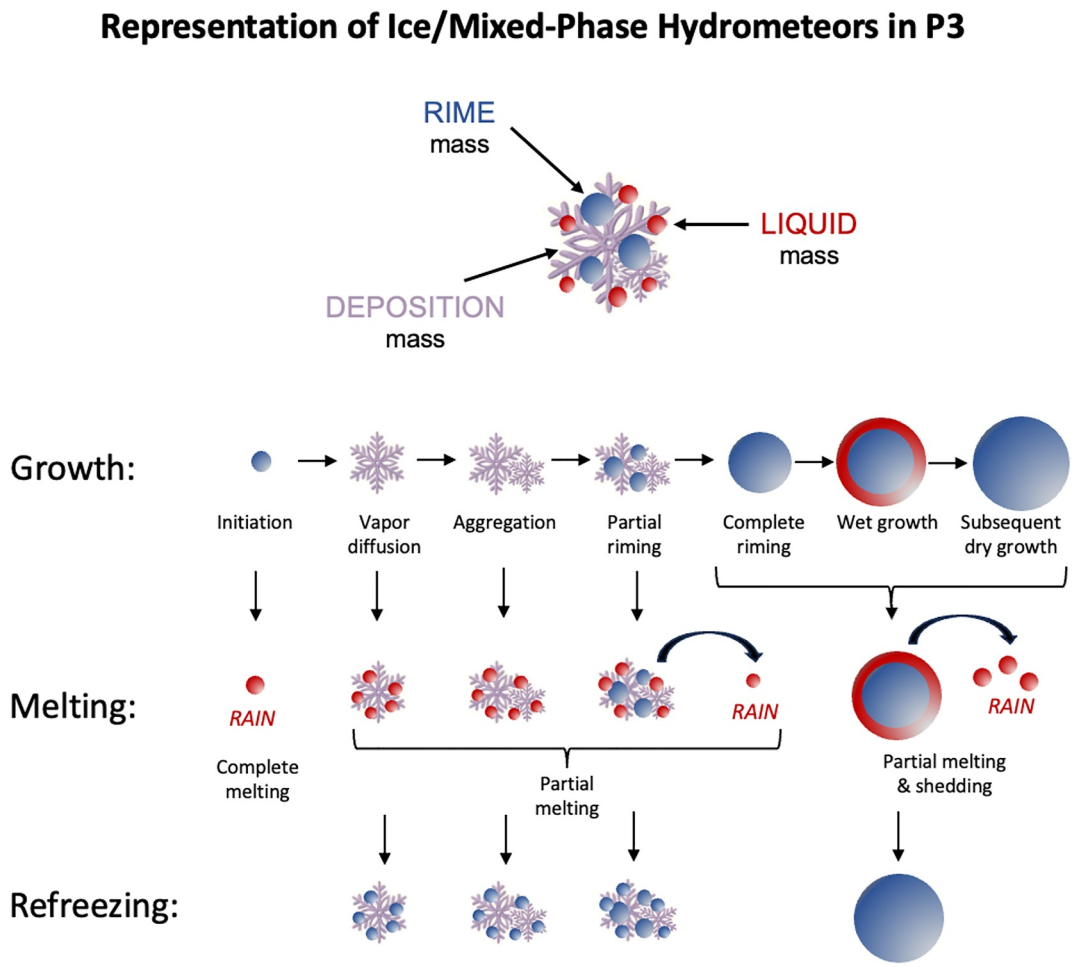


Figure 1. Depiction of the representation of ice-phase and mixed-phase hydrometeors and the main microphysical processes of growth, melting, and refreezing in the P3 scheme.

2.2. Summary of Idealized Simulations

The case chosen, while idealized, is based on a real hailstorm case from Alberta (Canada) on 13 June 2020. This case had a relatively cold cloud base, implying more likelihood of graupel embryos to frozen drop embryos (Young, 1993), relatively lower CAPE, yet the observed storm produced a prolific quantity of large hail (Joe et al., 2024). The initial sounding for the Alberta case had strong mid-to-upper-level vertical shear.

This study is done through examination of quasi-idealized cloud-resolving numerical simulations using the Cloud Model 1 (CM1) model (Bryan & Fritsch, 2002). CM1 is a nonhydrostatic fluid flow model, commonly used to simulate various atmospheric phenomena. We used the compressible dynamics option with sub-time steps for acoustic modes and fifth order advection of scalars using the Weighted Essential Non-Oscillatory scheme (Balsara & Shu, 2000). All simulations used a 200-m isotropic grid spacing and open lateral boundary conditions. The domain size was 1,000 × 1,000 horizontal grid points with 100 vertical levels (therefore with a 20 km model top). The model time step was 2 s. A Rayleigh damping layer was applied to the uppermost 5 km of the model with a damping timescale of 1/300 s^{-1} toward the base state sounding. The upper and lower boundaries were free slip and surface heat fluxes were zero. Horizontally homogeneous initial conditions were used, based on the single initial sounding. Subgrid-scale mixing used the 1.5 order prognostic turbulent kinetic energy method similar to Deardorff (1980). Radiation and all other physical processes besides microphysics were neglected. Convection was initiated using the vertical motion nudging method of Naylor and Gilmore (2012), with an ellipsoidal region (with a horizontal radius of 15 km and a vertical radius of 1,500 m) of vertical motion (peaking at 7.5 m s^{-1} in the center), with an inverse e-folding time of 0.5 s^{-1}, and nudging decreasing to zero over 20 min. A baseline

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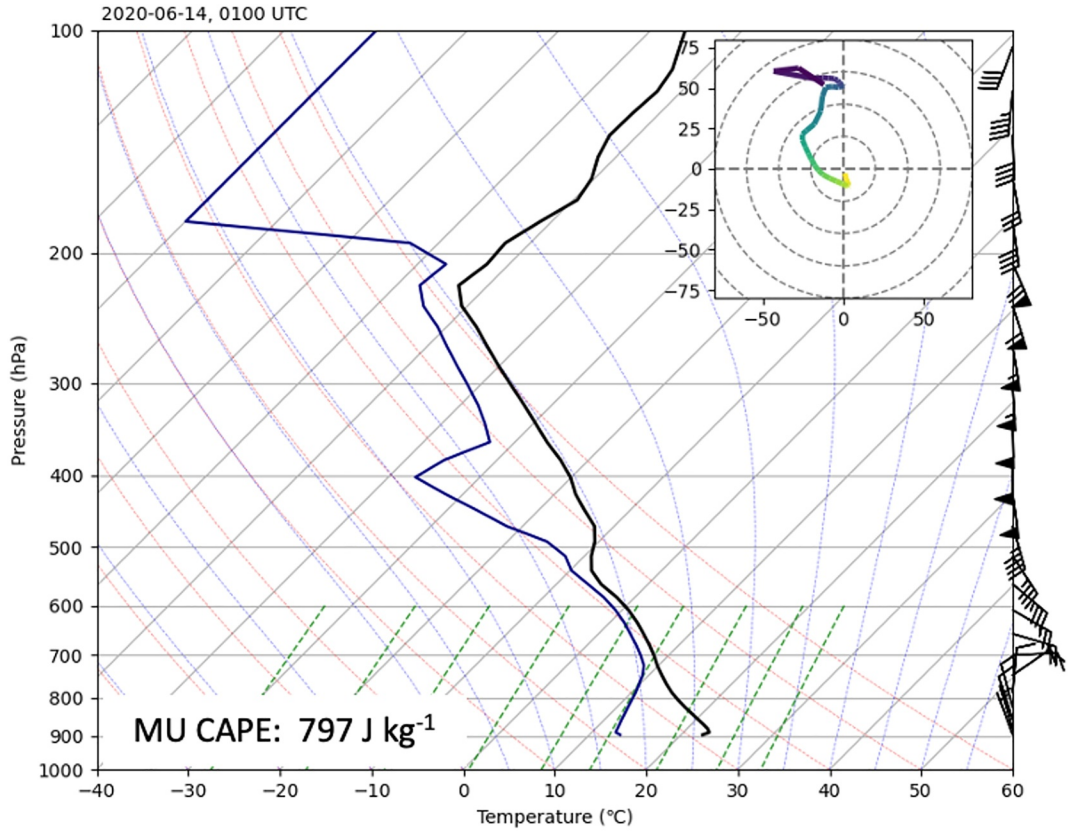


Figure 2. Initial sounding and hodograph, valid at 01:00 UTC 14 June 2020. See main text for additional details. Winds are in knots with conventional meaning for the barbs. The value of the most unstable CAPE is indicated. The sounding plot was created using MetPy.

configuration was run along with sensitivity tests related to prediction of liquid fraction and the number of ice-phase categories. All simulations examined in this study are listed in Table 1.

The sounding used to initiate each run, shown in Figure 2, was taken from a 6-hr model forecast sounding from ECCC's operational Regional Deterministic Prediction System (RDPS), valid at 00:00 UTC 14 June 2020, near the city of Calgary, Alberta, near the initiation region of the storm. The temperature and humidity profile was modified slightly to prevent initial instabilities. Also, the mean 0–6 km wind (the approximate storm translation speed) was subtracted from the sounding winds such that the model storms remained approximately in the domain center throughout the simulations. While the simulations in this study are idealized, we remark that the reflectivity structure of the baseline simulation was reasonably similar to the observed low-level reflectivity from a nearby radar (not shown).

3. Results of Baseline Simulation

3.1. Choice of Analysis Time

In order to reduce the quantity of analysis presented, a specific time was chosen for which the microphysical fields are examined. Periods of the modeled storm evolution of the baseline simulation (BASE) are shown in Figure 3, which shows snapshots at various times for the column-maximum model equivalent reflectivity ($Z_{e,c}$) and ice mixing ratios ($Q_{i,\mathrm{tot}}$) at the lowest model level, as well as the domain-maximum vertical motion. In order to reduce the quantity of analysis presented, a specific simulation time was chosen for which the microphysical fields are examined. The time of interest chosen was 80 min (Figure 3c), which is during the mature, quasi-steady stages of the model storm; it is representative of the general hail growth at other times during the mature stage. At this time, the storms is well developed and produce hail at the surface. In the figures discussed below, vertical cross sections are shown along the line in Figure 3c and similar locations for other simulations. The cross-section lines are

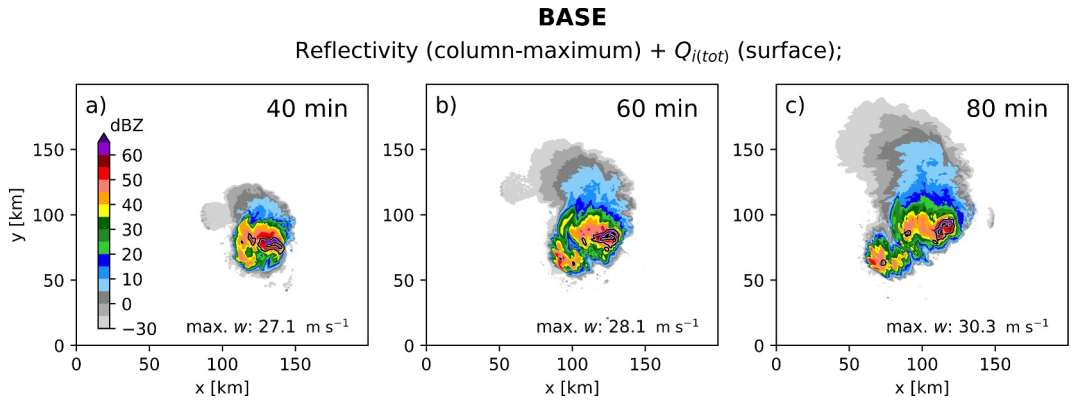


Figure 3. Column-maximum reflectivity ($Z_{e,c}$, shaded) and near-surface (lowest model level) ice mass mixing ratio ($Q_{i,\mathrm{tot}}$, contours of 10^{-6}, 0.1, 1.0 … g kg^{-1}) at 40, 60, and 80 min for BASE. Domain-maximum vertical motion (max. w) is indicated numerically in each panel. The arrow in panel (c) indicates the location of vertical cross sections in subsequent plots. (Note, the $Q_{i,\mathrm{tot}}$ contours are intended only to illustrate the location of hail near the surface.)

chosen subjectively (but consistently for each run) by the following criteria: the lines cross through the point of maximum $Q_{i,\mathrm{tot}}$ at the lowest model level (or the point of maximum $Z_{e,c}$ if the near-surface ice field is zero) and at prescribed meteorological angle 115°, from the rear to the front of the storms. For the remainder of the paper, microphysical fields are examined at these times and locations. However, the microphysical sensitivities discussed below are general and occur at other times (with occasional minor differences).

3.2. Overview of Baseline Simulation (BASE)

Results for the BASE run at 80 min are summarized in Figure 4. The left column shows vertical cross sections (see location in Figure 4c) of the total ice, cloud water, and rain mass contents (Figure 4a) and Z_e (Figure 4b), the near-surface (first model level) Z_e and ice mass content (Figure 4c) and the total and solid precipitation rates (Figure 4d). The panels in the right column shows various fields related to ice: the total ice mass content (Q_i) (which includes the liquid portion, which is relevant for other simulations), number concentration (N_i), rime mass fraction (F_{rim}), liquid fraction (F_{liq}; not relevant for the BASE but plotted for consistency with subsequent figures), mass-weighted mean density (ρ_i), mass-weighted mean fall speed (V_m), mass-weighted mean diameter (D_m), and maximum hail size ($D_{h,\mathrm{max}}$).

Despite this run having only a single ice-phase category, overall distributions of the ice physical properties are consistent with what one would expect in a strong deep convective storm. Most of the ice is heavily rimed ($F_{\mathrm{rim}} > 0.7$; Figure 4g), even the ice falling out of the anvil, owing to it originating within the strong convective updraft, with copious amount of liquid water available for riming (Figure 4a), before being lofted and transported horizontally. In the convective core the ice has high density (Figure 4i), large sizes (Figure 4k), and fast fall speeds (Figure 4j). At high altitudes (e.g., $z > 10$ km) in the anvil, the ice is small (Figure 4k) and dense (Figure 4i), with evidence of gravitational size sorting in forward part of the anvil, and larger mean ice sizes and fall speeds at lower elevations (Figures 4j and 4k). Note that frozen drops are initialized as rimed ice, in order to prescribe an initial bulk density as that of solid ice upon initiation. Technically this is not rime, but it allows the model to properly initiate and track the properties (density, shape). The maximum hail size, $D_{h,\mathrm{max}}$, is computed as the maximum physically reasonable observable size, estimated from the tail of the ice size distribution following the method described in Milbrandt et al. (2021), further masked to include only ice with density greater than 500 kg m^{-3} and a rime fraction greater than 0.8. $D_{h,\mathrm{max}}$, is indicated by the shaded regions in Figure 4l. The largest sizes are constrained to a narrow shaft in the convective core, reaching the surface along the right flank (Figures 4c and 4d), while smaller hail is lofted and ejected into the anvil, melting before reaching the surface. With the liquid-fraction shut off for the control run, the F_{liq} field (Figure 4h) is not applicable (or could be considered zero by design).

The accumulated total and solid (hail) precipitation all other runs is shown in Figure 5. The time period is from simulation time 45–90 min, thereby excluding most of the spin-up period (see Figure 3). The domain-mean values are indicated in the lower part of each panel. Note that these plots do not depict precipitation and hail swaths per se since the estimated storm translation vector (mean 0–6 km winds) was subtracted from the winds in the initial

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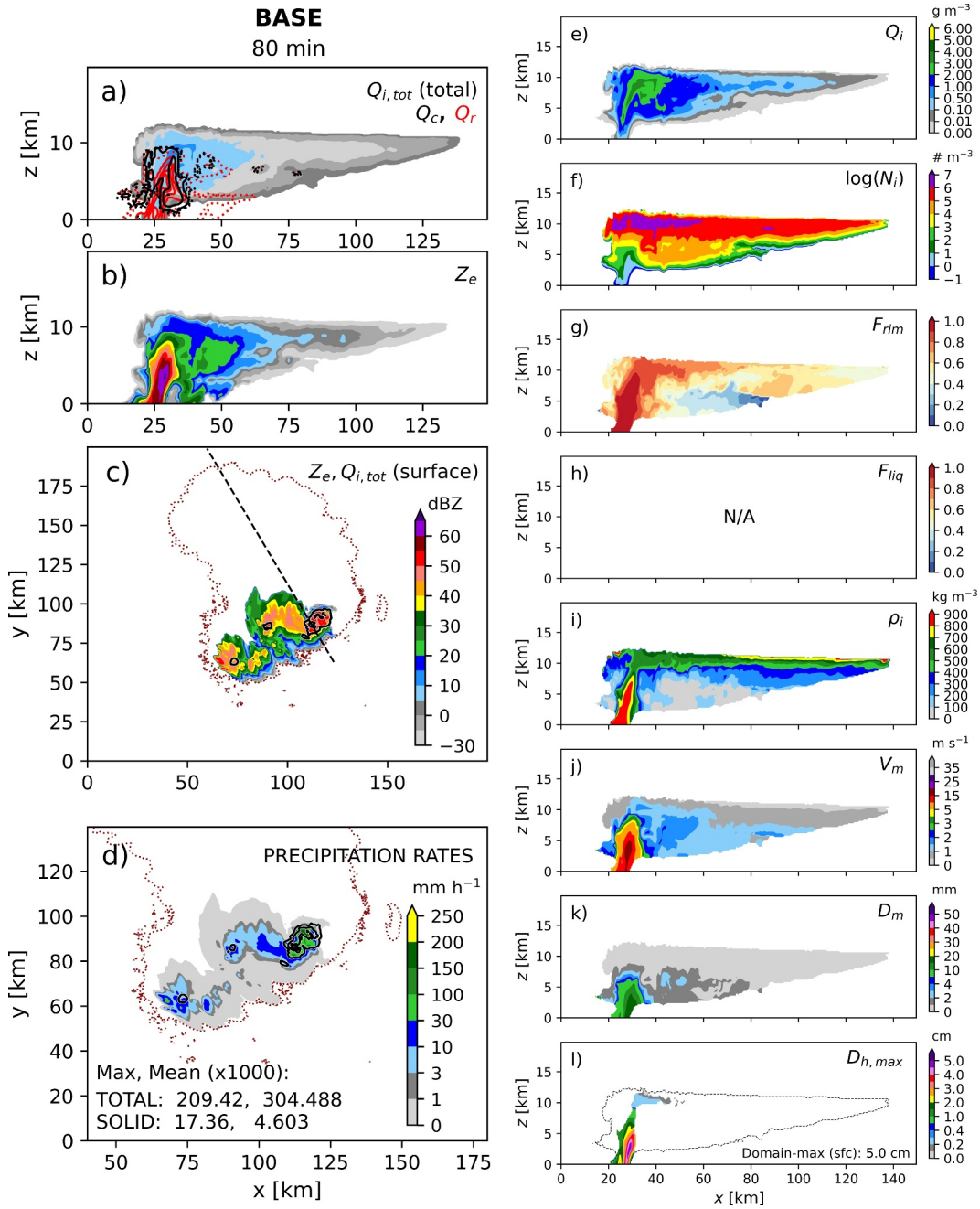


Figure 4. BASE simulation at 80 min. (a) Vertical cross-sections of Q_i (shading), Q_c (black contours), and Q_r (red contours), with shading boundaries and contours of 10^{-6}, 10^{-3}, 10^{-1}, 1, and 3 g m^{-3}; the first two contours for Q_c and Q_r are dotted. (b) Near-surface (lowest model level) Z_e (shading) and Q_i (contours; 10^{-6}, 10^{-1}, and 1 g m^{-3}). (c) Total (shaded) and solid (contours) surface precipitation rates (same shading boundaries/contours). Domain maximum and mean for each are indicated numerically. The right column has vertical cross sections of (e) Q_i, (f) N_i, (g) $F_{\rm rim}$, (h) $F_{\rm liq}$, (i) ρ_i, (j) V_m, (k) D_m, and (l) $D_{h,\rm max}$. Note, all mass and number mixing ratios (quantity kg^{-1}) are converted to concentrations (quantity m^{-3}) for plotting. The dotted red contours in panels (c) and (d) are the −30 dBZ $Z_{e,c}$ isopleths. The dotted line in panel (l) denotes the 10^{-12} g m^{-3} Q_i isopleth; the number in the right corner is the domain-maximum value of $D_{h,\rm max}$ at the surface. The line of the cross-sections for (a) and (b), through the peak surface Z_e, is indicated in panel (c); the cross-sections for (c–l) are shifted slightly to the right, crossing through the peak surface Q_i.

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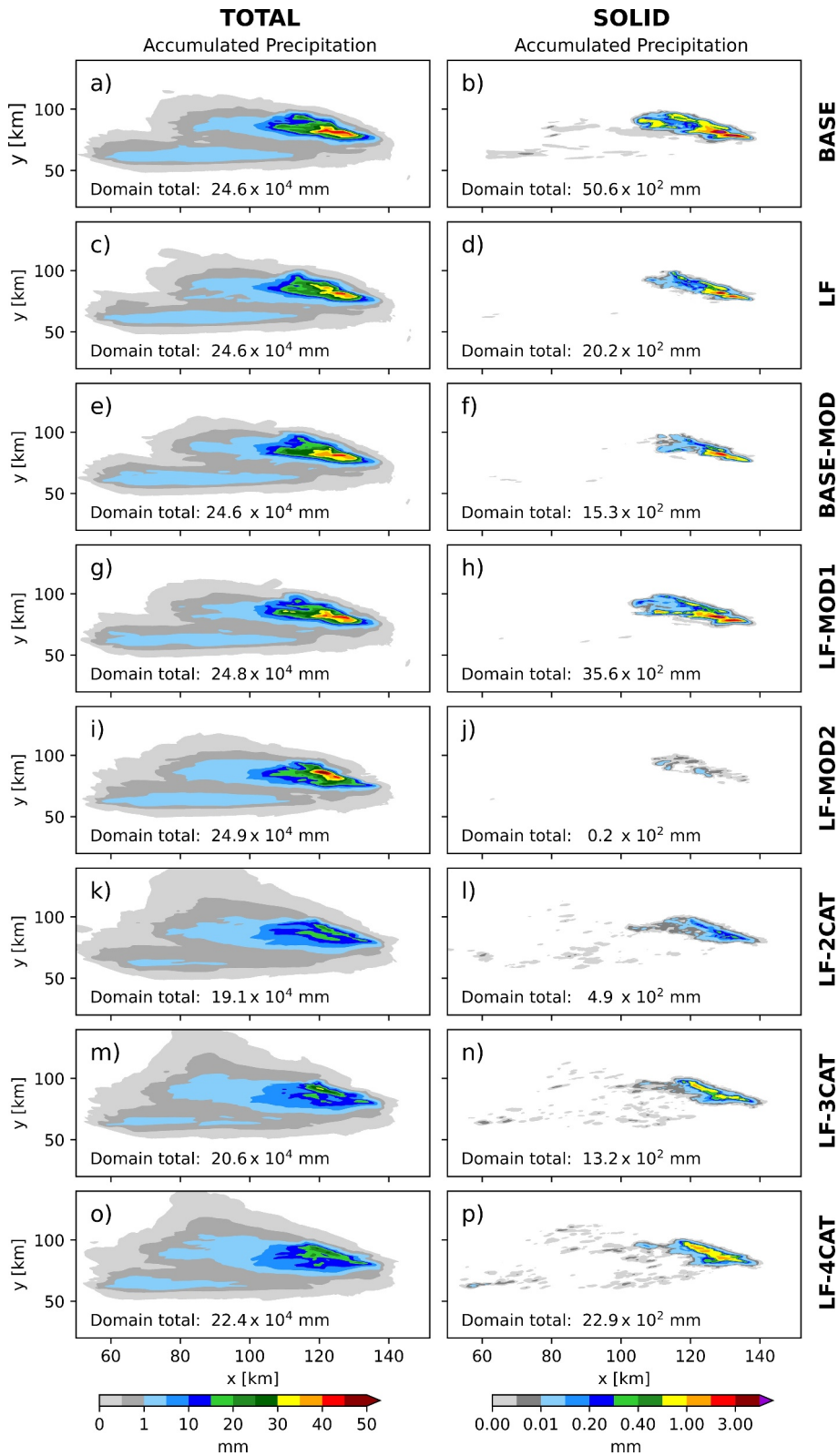


Figure 5. Accumulated total (left) and solid (right) precipitation (45-min, from integration times 45–90 min) for each simulation. The domain-total quantities are indicated in the bottom of each panel.

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sounding. The precipitation cores, that is, the regions of high precipitation rates (e.g., Figure 4d), still translate somewhat over time toward the "west/north-west" creating apparent tracks, but there is no particular significance to their dimensions or locations. Nevertheless, the precipitation plots are done the same for each simulation, thus Figure 5 can be used to compare the precipitation and hailfall at the surface from the various sensitivity runs. Note that comparison of accumulated precipitation is preferable for this purpose to precipitation rates at a snapshot in time since instantaneous rates can be intermittent for hailstorms. Also, note that the fact that these are not strict precipitation swaths likely exaggerates the impacts of the precipitation differences compared to what would be shown in a true ground-relative framework.

4. Microphysics Sensitivity Tests

4.1. Impacts of Liquid Fraction

The ice fields for the simulation with the predicted liquid fraction activated, LF, are summarized in Figure 6. Most of the ice fields are quite similar to those of BASE (Figure 4), but with some notable and important differences. The $F_{\rm liq}$ field is now defined and while it is zero for most locations with ice present, it increases to 1 (Figure 6h) with decreasing height as the ice melts to rain (Figure 6a). Although it is not apparent in here, $F_{\rm liq}$ can also be non-zero in the hail growth zone if the ice is undergoing wet growth. Also, with the predicted liquid fraction the model reflectivity is more realistic in melting regions of stratiform weather systems as a result of the bright band (not shown) being accounted for (Cholette et al., 2023). The important impact of the inclusion of liquid fraction pertaining to this study is that the quantity and size of ice/hail reaching the surface is notably reduced (Figures 6d, 6e, and 6l). This is consistent with reduced domain-mean solid precipitation rates (Figure 5d).

This significant reduction of hail reaching the surface is due to differences in the way melted ice mass is treated with liquid fraction on and the closure assumptions pertaining to the ice number tendency during melting and shedding, all of which increase the realism of these parameterized processes in P3. With liquid fraction off, melting in P3 is treated such that the amount of total ice mass that melts in one time step is computed, following the heat balance equation (see Morrison and Milbrandt (2015) for details), and is assumed to be immediately and completely shed to rain. The $N_{i,\rm tot}$-tendency due to melting is parameterized using the closure assumption of a constant mean-mass diameter D_{mm}, defined by

$$D_{mm} = \left(\frac{6}{\pi\rho_i}\cdot\frac{Q}{N}\right)^{1/3}, \tag{2}$$

which is commonly used in multi-moment bulk schemes (e.g., Ferrier, 1994; Mansell et al., 2010; Milbrandt & Yau, 2005b; Morrison et al., 2005; Ziegler, 1985). In Equation 2 and the equations below, the "*i,tot*" subscripts are excluded from $Q_{i,\rm tot}$ and $N_{i,\rm tot}$ for conciseness. Note that for a gamma size distribution, this is equivalent to assuming a constant value of $(\mu + 1)/\lambda$ for this process (or a constant λ for an inverse-exponential distribution, with $\mu = 0$). With this assumption, the ratio $Q_{i,\rm tot}/N_{i,\rm tot}$ remains constant so $N_{i,\rm tot}$ decreases proportionally to $Q_{i,\rm tot}$ during melting, following

$$\left(\frac{dN}{dt}\right)_{\rm MELT} = \frac{N}{Q}\cdot\left(\frac{dQ}{dt}\right)_{\rm MELT}. \tag{3}$$

where $(dQ/dt)_{\rm MELT}$ is the melting rate of $Q_{i,\rm tot}$. With liquid fraction on, the treatment of the melted ice mass in P3 is physically more realistic (see Cholette et al. (2019) for details). In a given time step during melting, a portion of the solid mass, $Q_{i,\rm tot} - Q_{i,\rm liq}$ (i.e., deposition plus rime mass), melts to liquid. Small ice particles are assumed to melt completely in the time step and thus the mass goes directly to rain (Q_r), with $N_{i,\rm tot}$ decreasing as per Equation 3 for this portion of the melting (only). The rest of the melted ice mass, however, is converted from solid to liquid but remains within the mixed-phase particles ($Q_{i,\rm rim}$ decreases, $Q_{i,\rm liq}$ increases, and $Q_{i,\rm tot}$ remains the same), increasing the liquid fraction locally and with no change to $N_{i,\rm tot}$. Shedding of liquid on mixed-phase particles is then calculated explicitly (see Cholette et al., 2019, 2023) and depends on the rime mass fraction, where at the extremes unrimed ice does not shed whereas fully-rimed ice sheds at the maximum rate. During shedding, $N_{i,\rm tot}$ does not change while the rain mass and number increase. As a result, with liquid fraction on the

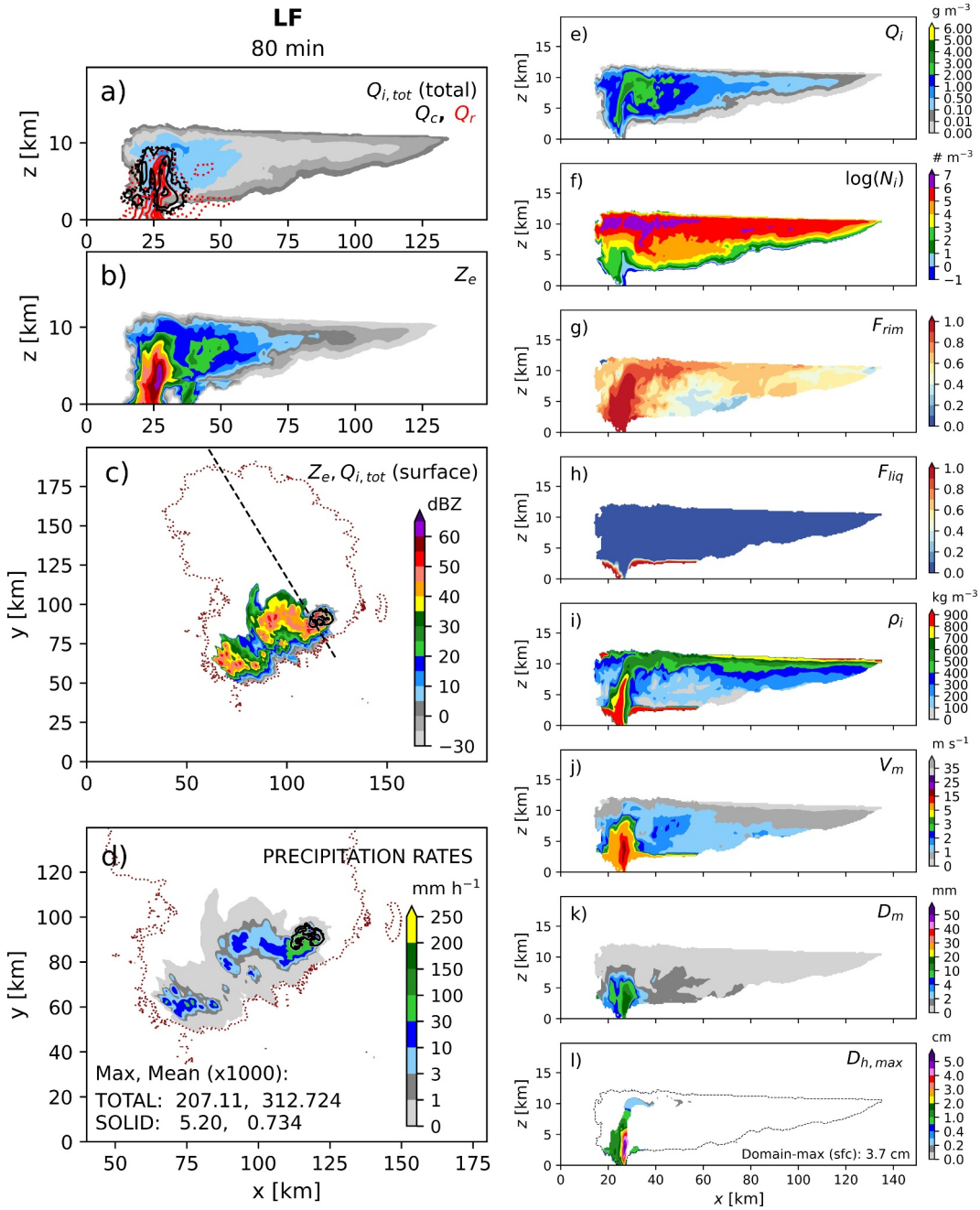


Figure 6. As in Figure 4 but for LF.

melting of heavily rimed ice combined with shedding tends to result in a decrease in the mean ice size, as per Equation 2.

For unrimed or lightly-rimed ice, the differences in mass changes during melting and the impacts of this are significant since with liquid fraction on the melted mass remains in mixed-phase particles, increasing their fall speed toward that of rain until the ice melts completely. For heavily-rimed ice, although there is a temporary transfer of most of the melted mass to liquid within mixed-phase particles, most of this mass is quickly shed; thus, the mass transfer from solid ice to rain is similar to that with liquid fraction off. However, the difference in the treatment of $N_{i,\mathrm{tot}}$ during melting/shedding results in a much different change in D_m, with the size decreasing

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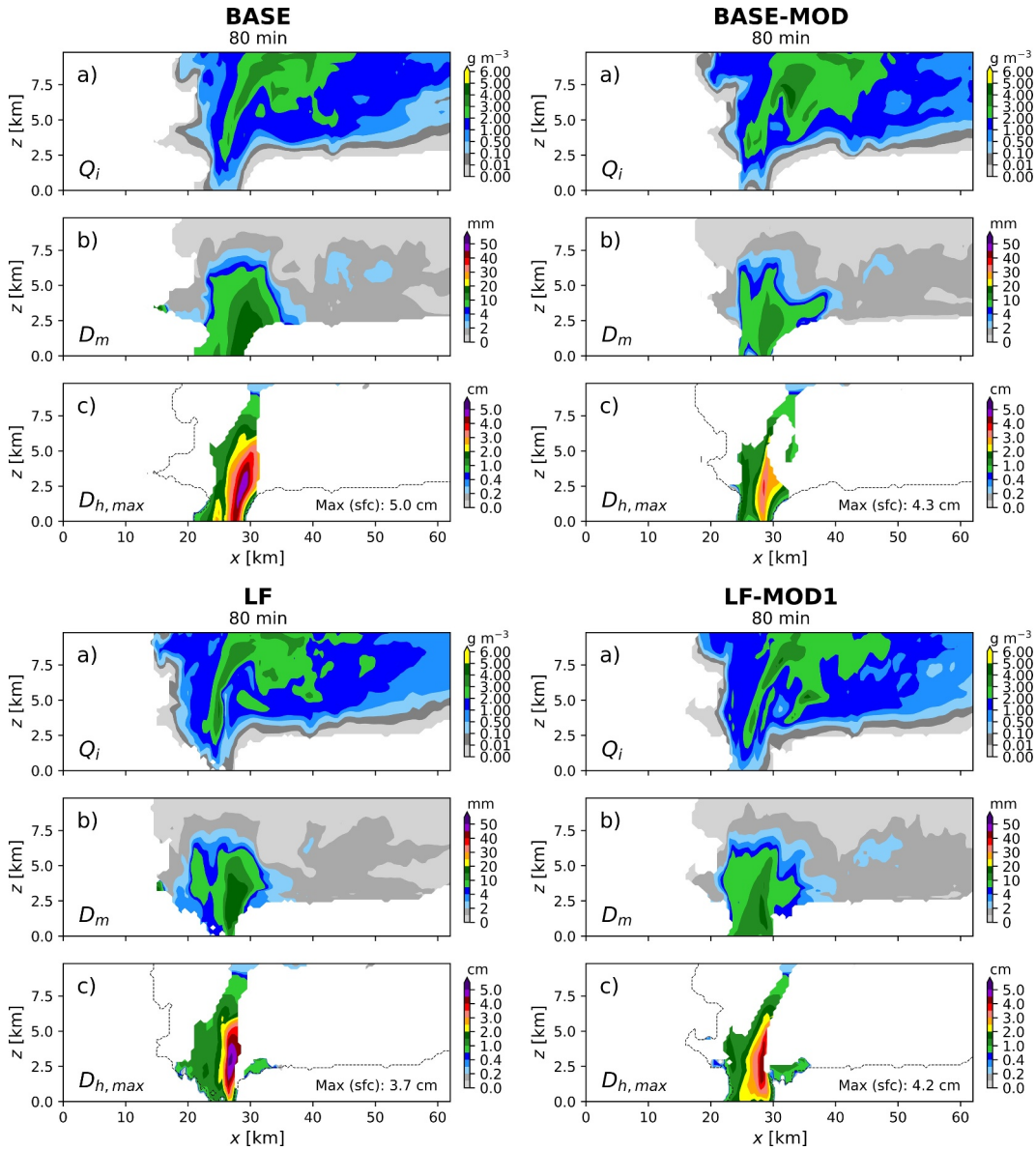


Figure 7. Vertical cross-sections of Q_i, D_m, and $D_{h,\max}$ at 80 min for BASE [top left; (a–c)], BASE-MOD [top right; (d–f)], LF [bottom left; (g–i)], and LF-MOD1 [bottom right; (j–l)]. The dotted lines in (c, g, j, and m) denotes the 10^{-12} g m^{-3} Q_i isopleth.

much faster with liquid fraction on. This results in slower fall speeds, greater total surface area (since a given mass is distributed over more particles), and thus a faster mass melting rate—and ultimately melting hail is more likely to completely melt before reaching the surface. Therefore, with the exception of the decrease in $N_{i,\mathrm{tot}}$ that occurs as the small melted fraction transferred directly to rain, for the melting that is first retained in the liquid portion of the mixed-phase particles and then shed, $N_{i,\mathrm{tot}}$ remains constant.

To illustrate that the differences described above are indeed responsible for the reduced hail at the surface in the LF simulation with liquid fraction on, two sensitivity tests were run. For the first test (BASE-MOD), the code in the BASE configuration was modified such that $(dN/dt)_{\mathrm{MELT}}$ is set to zero, similar to the overall effect with liquid fraction on in LF (except for the portion of small ice melting directly to rain). In the second test (LF-MOD1), the LF configuration was modified such that $(dN/dt)_{\mathrm{MELT}}$ is computed by Equation 3 following the closure assumption for melting in BASE, with $(dN/dt)_{\mathrm{SHED}}$ set to zero since it is implicitly included in melting. The

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impacts on ice mass and mean sizes near the surface are shown in Figure 7 (with fields from the unmodified BASE and LF from Figures 4 and 6 repeated and zoomed in to facilitate comparison). The BASE-MOD run (Figures 7h, 7j, and 7l) is now similar to the LF run (Figures 7e–7g), with a reduction in ice mass reaching the surface. Both D_m and $D_{h,\max}$ are reduced (Figures 7i and 7j) as is the solid precipitation reaching Figure 5f. Conversely, LF-MOD1 (Figures 7k–7m) is similar to BASE (Figures 7a–7c), with more ice/hail reaching the surface (Figure 7k), increased particle sizes at the surface (Figures 7l and 7m), and solid precipitation (Figure 5h) now closer to that of BASE (Figure 5b). Clearly, therefore, the differences in the closure assumptions for the changes in $N_{i,\mathrm{tot}}$ during melting/shedding are largely responsible for the notable reduction of hail at the surface with the liquid fraction on configuration.

To illustrate further the differences in melting between these configurations, process rates related to melting/shedding for LF are shown in Figure 8. Here melting is partitioned into the portion of frozen ice mass that melts directly to rain (MELT1) and the portion that transfers frozen mass to liquid mass in the mixed-phase particle (MELT2). Figures 8a and 8b shows $(dQ/dt)_{\mathrm{MELT1}}$ and $(dN/dt)_{\mathrm{MELT1}}$, respectively. The mean particle size due this sub-process remains constant. Figure 8c shows $(dQ/dt)_{\mathrm{MELT2}}$, which does not result in any change in $Q_{i,\mathrm{tot}}$ (only a transfer from ice to liquid that remains on the particle), with the mass loss rate due to shedding given by $(dQ/dt)_{\mathrm{SHED}}$ (Figure 8d). The resulting rate of change of D_{mm} due to the combined melting and shedding, $(dD/dt)_{\mathrm{MELT\,+\,SHED}}$, is given by

$$\left(\frac{dD}{dt}\right)_{\mathrm{MELT+SHED}} = \frac{1}{3}\left(\frac{\pi}{6}\rho_i\right)^{-\frac{1}{3}} \cdot \left[Q^{-\frac{2}{3}}N^{-\frac{1}{3}}\left(\left(\frac{dQ}{dt}\right)_{\mathrm{MELT1}} + \left(\frac{dQ}{dt}\right)_{\mathrm{SHED}}\right) - Q^{\frac{1}{3}}N^{-\frac{4}{3}}\left(\frac{dN}{dt}\right)_{\mathrm{MELT1}}\right], \tag{4}$$

and is shown in Figure 8e. Clearly the mean size of hail that melts and sheds in the convective cores ($x \sim 25$ km) decreases with the liquid fraction configuration. Note, in computing Equation 4 (which is only diagnostic, for Figure 8e), which is the derivative of Equation 2 with respect to time, ρ_i is held constant for simplicity, so in fact Equation 4 underestimates the rate of decrease in mean particle size. This is because for MELT2, although there is no change in total mass or number, frozen mass is transferred to liquid ($Q_{i,\mathrm{liq}}$) which increases the mean particle density, thereby decreasing the mean size as per Equation 2.

Given that the LF configuration indeed allows for a more realistic representation of melting and shedding, and also for representing complex mixed-phase winter precipitation (Cholette et al., 2019, 2020, 2024), a way to improve this in the liquid fraction-off configuration may simply be to replace the constant D_{mm} closure assumption for melting with the one tested in BASE-MOD, at least for large values of F_{rim}. This may be considered for future versions of P3 since there is some computational cost associated with the LF configuration (due to the additional prognostic variable, $Q_{i,\mathrm{liq}}$). Similarly, the $(dN/dt)_{\mathrm{MELT}} = 0$ approach may be a simple improvement in traditional bulk schemes for the graupel and hail categories that do not include prognostic liquid fraction. Although the new closure assumptions for melting/shedding are physically based and exploit the capability of predicting the liquid fraction, the results presented do not prove that these processes are indeed modeled optimally. Deeper investigation into the representation of melting/shedding in P3 could be done following the approach of Mansell et al. (2020) who proposed an alternative to mitigating the assumption of Equation 3 guided by detailed bin scheme calculations of hail melting.

Shedding also happens during wet growth which can, in principle, be modeled more realistically with the predicted liquid fraction. With liquid fraction off, the shedding rate (mass) is computed as the dry growth rate (i.e., the total accretion rate of liquid water) minus the wet growth rate, computed from a heat balance equation, based on the Shuman-Ludlam limit (Young, 1993). With liquid fraction on, this difference in unfrozen collected mass remains as liquid on the particles (F_{liq} increases); shedding of a portion of this liquid is then computed separately. This is depicted in Figure 1. Thus, wet hail can be represented and the liquid remaining can re-freeze onto the existing ice once wet growth stops, so hail should in principle be able to acquire larger mass contents and sizes in situations with alternating wet and dry growth. There is no evidence of wet hail at temperatures below 0°C in any of the simulations in this study (which would be identified with values of $F_{\mathrm{liq}} > 0$ in the appropriate regions, at elevations above 3 km; Figure 8). This is likely because the conditions for wet growth are never met for this relatively weak case (e.g., with peak updrafts around 30 m s^{-1}; Figure 2), though it is also possible that the shedding rate is too large in the microphysics scheme.

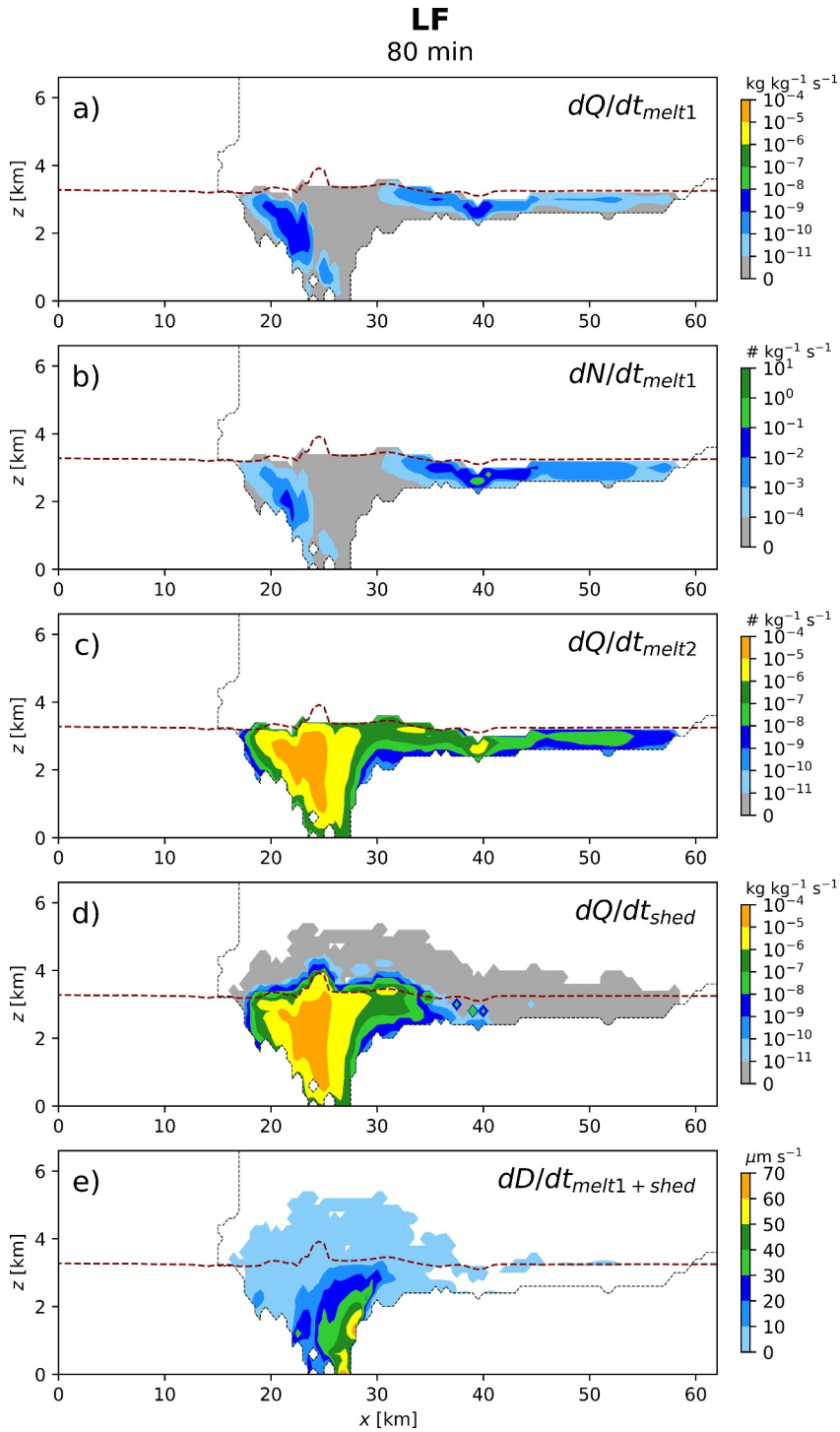


Figure 8. Process rates related to melting and shedding for LF at 80 min: (a) and (b) $Q_{i,\mathrm{tot}}$ and $N_{i,\mathrm{tot}}$ tendencies, respectively, due to complete melting (to rain) of small particles (MELT1); (c) Tendency for mass transfer from solid to liquid within mixed-phase particles due to melting (MELT2); (d) $Q_{i,\mathrm{tot}}$ tendency due to shedding; (e) D_m tendency due to combination of melting to rain and shedding calculated from Equation 4. See main text for details. Positive values denote decreases. The red-dashed line denotes the 0°C isotherm. The dotted lines denote the 10^{-12} g m^{-3} Q_i isopleth.

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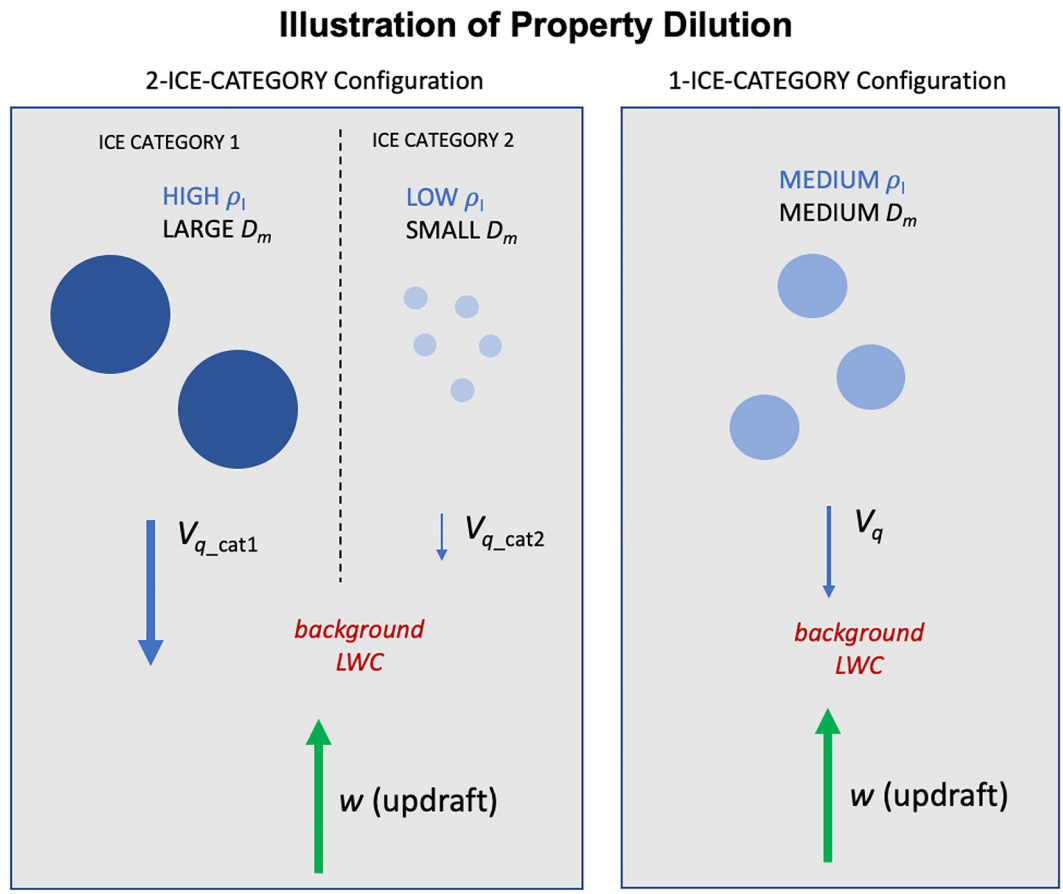


Figure 9. Conceptual illustration of property dilution using generic ice-phase categories. The circles depict ice particles, with the sizes proportional to the mean-mass diameters and shading intensity proportional to the bulk density.

The topic of wet hail growth is a potentially interesting aspect of the liquid-fraction configuration of P3 and will be examined in a future study.

4.2. Impacts of the Number of Ice Categories

In order to examine the impacts of multiple generic ice-phase categories, it is first necessary to understand the impacts of including SIP processes in the single-ice-category configuration and, more generally, the concept that we refer to as *property dilution*. Also, as illustrated below, it is necessary to establish a proper control simulation in order to study the impacts of multiple categories. Property dilution occurs when, for example, two modes—that is, sets of bulk physical properties—that could otherwise coexist at the same point in time and space are represented with a single distribution and hence set of bulk properties; the result is a kind of averaging of the properties, hence "property dilution." This is illustrated in the conceptual diagram in Figure 9. On the left side, two distinct modes of ice are represented (at the same location) with the two ice categories, one with large, high-density (and therefore a high bulk fall speed) ice and the other with small, low-density (and a low fall speed) ice, with a given background liquid water content and updraft. Because of the separation of properties, the large ice mode (category 1) would be able to remain in the hail growth zone longer due to near-matching bulk fall speeds and updraft speeds, and thus grow to larger sizes, while the slower sedimenting ice in category 2 would be quickly transported aloft. In contrast, if the ice on the left side of Figure 9 were to be represented with a single generic category, as depicted on the right side, the bulk ice properties would be a kind of average of the two categories on the left, with a medium bulk density, size, and fall speed, thereby removing the high and low. The result is that the ice in the single category configuration will have a smaller cross-sectional area and fall speed as the ice in category 1 of the two-category configuration and therefore will grow by accretion more slowly; ultimately this will result in less large hail than in the two-category configuration.

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The problem of property dilution for the simulation of hail is particularly egregious when parameterized SIP processes are included. Rime splintering, for example, produces numerous tiny ice crystals when graupel undergoes riming at temperatures between −3 and −8°C (Hallett & Mossop, 1974). With the concept of property dilution in mind, this would be expected to be detrimental to the simulation of large hail with the 1-ice-category configuration for the reasons discussed regarding the conceptual illustration above (Figure 9). For this reason, the default 1-category configuration of P3 has all SIP processes deactivated since the dilution problem is deemed (based on previous tests) to be more detrimental than the exclusion of these parametrized physical processes. Note, in the current version of P3, only Hallet-Mossop rime splintering is included for multi-ice-category configurations, though the effects of other SIP mechanisms in P3 have been explored (e.g., Huang et al., 2021; Lachapelle et al., 2024; Qu et al., 2022) and will be considered for the "official" version of the scheme in the future.

Despite the default one-category configuration with SIP off, in order to examine properly the impacts of increasing the number of generic ice-phase categories (only) for the simulation of hail, a proper one-category control simulation must be established, with all other aspects being identical, including the treatment of SIP. Therefore the one-category LF simulation was re-run with the rime-splintering process activated; the results of this simulation (LF-MOD2) at 80 min are shown in Figure 10. There are several important differences between this run and the LF (Figure 6). The increase in the total ice number concentration in the convective region (Figure 10f) is clearly evident upon comparison to LF (Figure 6f). This in turn leads to smaller mean sizes (Figure 10k) and fall speeds (Figure 10j) even though the total ice mass in the convective region is larger, at least aloft (Figure 10e). Consequently, the reflectivity in the core is also greatly reduced (Figure 10b). Overall, the quantity of hail reaching the surface is greatly reduced, as is evident from the accumulated solid precipitation (Figure 5j) and the instantaneously ice mass reaching the surface at 80 min (Figure 10e), as is the maximum hail sizes at the surface in the convective core (Figure 10l). The LF-MOD2 simulation is probably not as realistic as the LF run, which indeed is why SIP is deactivated for the default one-ice-category configuration of P3, but it is the appropriate configuration to serve as the control run to examine the impacts of the number of ice categories.

As illustrated in Section 3.2, even with a single generic ice category P3 can simulate a wide and continuous range of physical properties of ice-phase particles, and hence a wide range of frozen hydrometeor types. With the liquid fraction configuration, mixed-phase particles can also be represented and the associated microphysical processes better modeled. The limitation, however, is that only a single set of properties—and hence one dominant particle type—can be present at a given point in time and space, as discussed in the previous section. The multi-category configuration of P3 removes this limitation and allows for multiple sets of properties, and thus multiple modes, to coexist and to evolve independently. The details of ice initiation into a given category and the merging of categories with similar properties are explained in Milbrandt and Morrison (2016).

To examine the impact of multiple categories on the simulation of hail, a set of sensitivity simulations was done, increasing the number of categories (*nCat*) from 1 to 4, incrementally. All of these runs were with predicted liquid fraction and SIP activated (Table 1), thus the appropriate control simulation for this set of tests is LF-MOD2. The fields for each of the ice categories in LF-2CAT are shown in the middle and right columns of Figure 11. At any given location, the physical properties can be quite different between the two categories and also different from the single-category run, LF_MOD2 (Figure 10). With the 3-category run (LF-3CAT, Figure 12), not only is the ice in category 3 (Figure 12, right column) different from the other two, but the ice in categories 1 and 2 of LF-3CAT (Figure 12, left and middle columns) are different from those of LF-2CAT. For example, in category 2 of LF-3CAT there is much higher number concentration of ice aloft downwind of the convective core (Figure 12n) compared to LF-2CAT (Figure 11n) and there is more hail at mid-levels in the convective core (Figure 11t). These and other differences emphasize that a given ice category in the multi-category configuration is not predisposed to represent a particular mode/type of ice; each category is equally continuous and variable. Similarly, a given category is not predisposed to a particular set of dominant properties, though for both LF-2CAT and LF-3CAT most of the ice mass resides in category 1 (top rows of Figures 11 and 12). The fields for the various ice category of the LF-4CAT run (not shown) support this further. Also, the plots for LF-4CAT (not shown) corresponding to the left columns of Figures 4, 6, 9, 11, and 12 are very similar to those of LF-3CAT (Figure 12).

Figure 13 summarizes the "bulk" ice fields for all categories for the LF-2CAT, LF-3CAT, and LF-4CAT simulations. For the total mass and number this is the sum of $Q_{i,\mathrm{tot}}(i)$ and $N_{i,\mathrm{tot}}(i)$ over all i in 1 to $nCat$ (converted to mass and number concentrations, Q_i and N_i); the "total" F_{rim} and F_{liq} are computed from the appropriate sums of

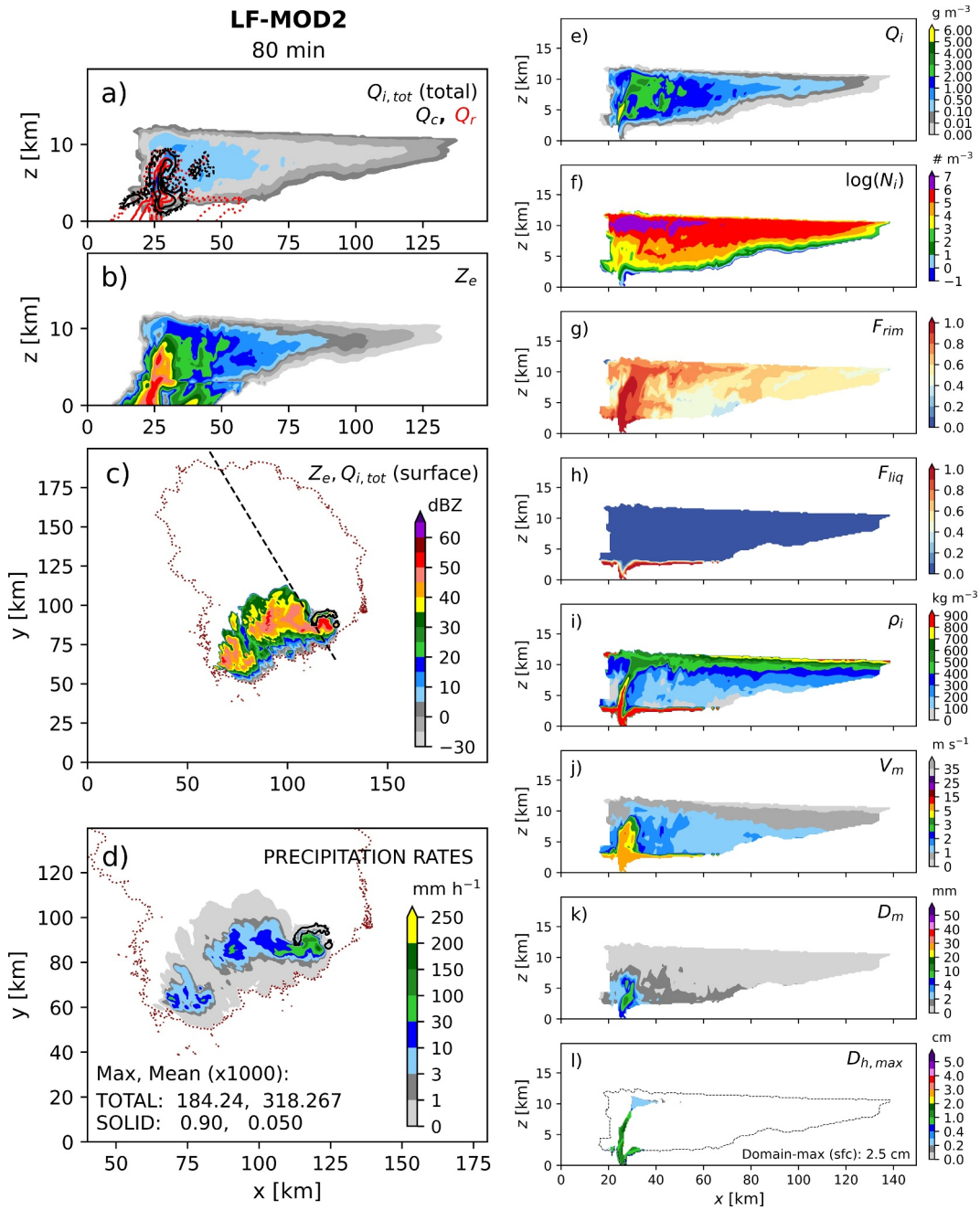


Figure 10. As in Figure 4 but for LF-MOD2 run.

the ice fields; ρ_i, V_m, and D_m are mass-weighted averages of each category; and D_{h_max} is the maximum value amongst the categories. These bulk fields can be compared directly to the 1-category control run, LF-MOD2 (Figure 10). Despite the internal variability within the individual ice categories exhibited in Figures 11 and 12 and the 4-category run (not shown), the bulk fields (Figure 13) are broadly similar for all of the simulations, as well as to those of the 1-category control run (LF-MOD2, Figure 10).

However, there are some important differences in the simulations with the variation of the number of ice categories. In general, the hail quantity and size increase as the number of ice categories is increased. This is evident from the hail mass reaching the surface around the convective core ($x \sim 25$ km), the mean particle sizes D_m, the maximum hail sizes $D_{h,\max}$, as well as the domain-mean solid precipitation accumulation (Figure 5). As the

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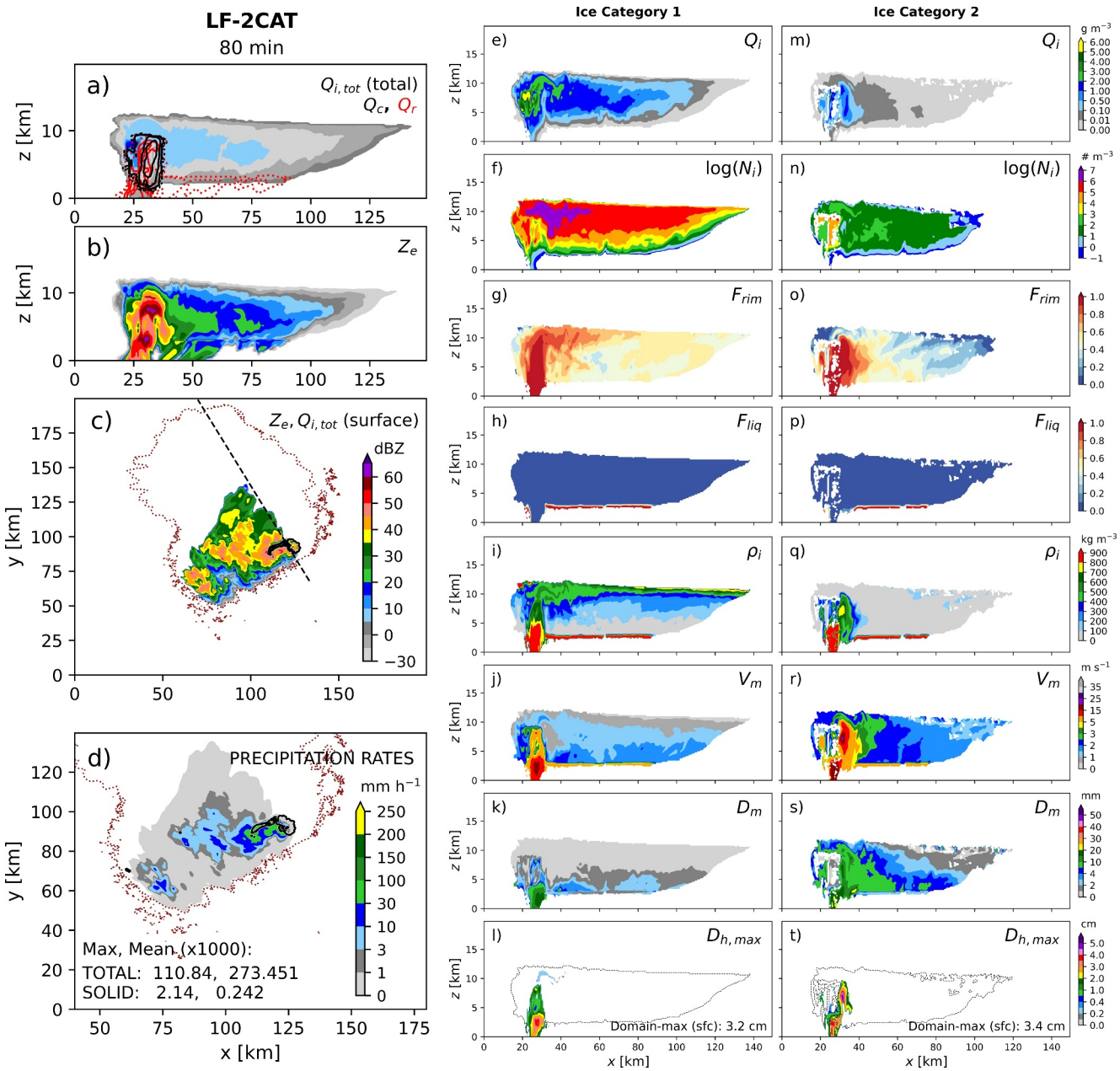


Figure 11. As in the right column of Figure 4 but for ice category 1 (middle) and category 2 (right) for OK-LF-2CAT.

number of ice categories increases, however, the differences in the overall ice properties, including those related to hail, appears to decrease with each additional ice category. For example, the overall differences between LF-2CAT and LF-3CAT (Figure 13) are smaller than the differences between LF-2CAT (Figure 13) and LF-MOD2 (Figure 11), and in turn the differences between LF-3CAT and LF-4CAT (Figure 13) are even smaller. There is, however, a notably increase in the solid precipitation between LF-3CAT and LF-4CAT (Figure 5). Nevertheless, the overall solution appears to be converging with an increasing number of ice categories. This is elaborated upon in the next section.

The total (and hence liquid) precipitation is also notably reduced with the move from one ice category (LF-MOD2; Figure 5i) to two (LF-2CAT; Figure 5k), both in the convective core region (by about half) and throughout the domain in general (by about 25%). With the increase in the number of ice categories to three (LF-3CAT; Figure 5m) and to four (LF-4CAT; Figure 5o), the total precipitation increases gradually. The changes in

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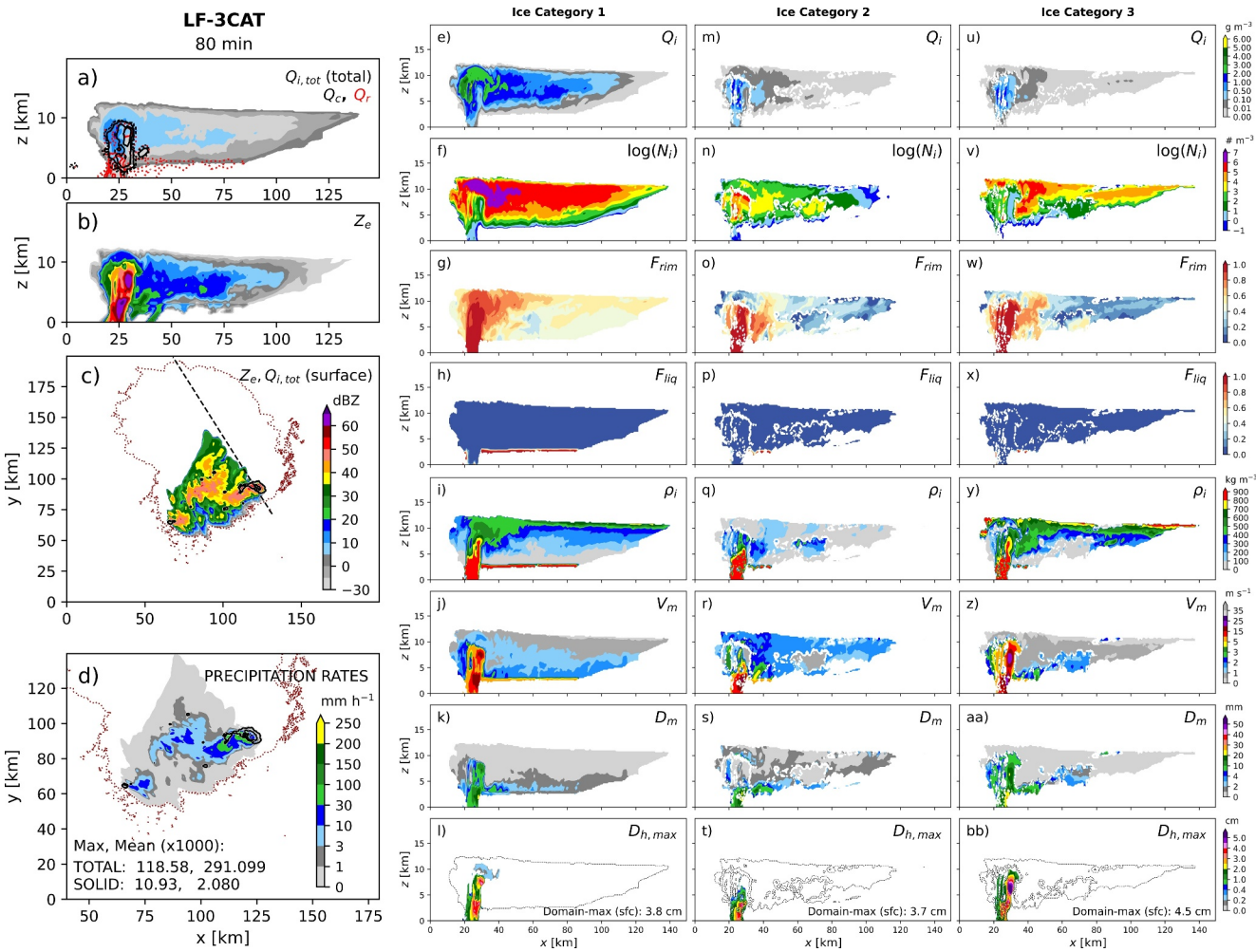


Figure 12. As in the right column of Figure 4 but for ice category 1 (column 2), category 2 (column 3), and category 3 (column 4) for LF-3CAT.

surface rainfall can be understood by consideration of the property dilution concept illustrated in Figure 9. With the activation of SIP in LF-MOD2 compared to BASE, the ice number concentration increases and thus mean ice sizes and fall speeds decrease. Although this prevents ice from reaching the surface as hail, most of the mass in the convective region still reaches the melting layer (around 3 km AGL) and ultimately reaches the surface as rain, thereby resulting in an overall similar total precipitation for LF-MOD2 (Figure 5i) and BASE (Figure 5a). However, with the introduction of a second ice category, the total accreted liquid mass by the ice in LF-2CAT is shared between the two ice categories, as depicted in Figure 9 but with much of the ice (category 2 in Figure 9; much of category 1 in Figure 11) having smaller mean sizes and fall speeds. This ice is thus unable to sediment through the updraft and ultimately to the surface (arriving as either as ice or rain) but rather it is transported aloft and into the anvil; therefore, the overall precipitation is reduced. As the number of ice categories is increased, the property dilution is reduced and more large ice can reach the surface, either as hail or melting to rain, thereby increasing the ice and liquid precipitation (Figures 5m–5p).

5. Discussion

Although the simulations presented are idealized and evaluation against a numerical "truth" solution is impossible since no analytical solution or perfect microphysics scheme exists (Morrison et al., 2020), it is reasonable to expect that the aspects of P3 explored above improve the overall realism in representing ice-phase and mixed-phase hydrometeors in general, including the modeling of hail. That does not, however, necessarily mean that for a real-case simulation (or prediction in an NWP model) with predicted liquid fraction and/or multiple ice

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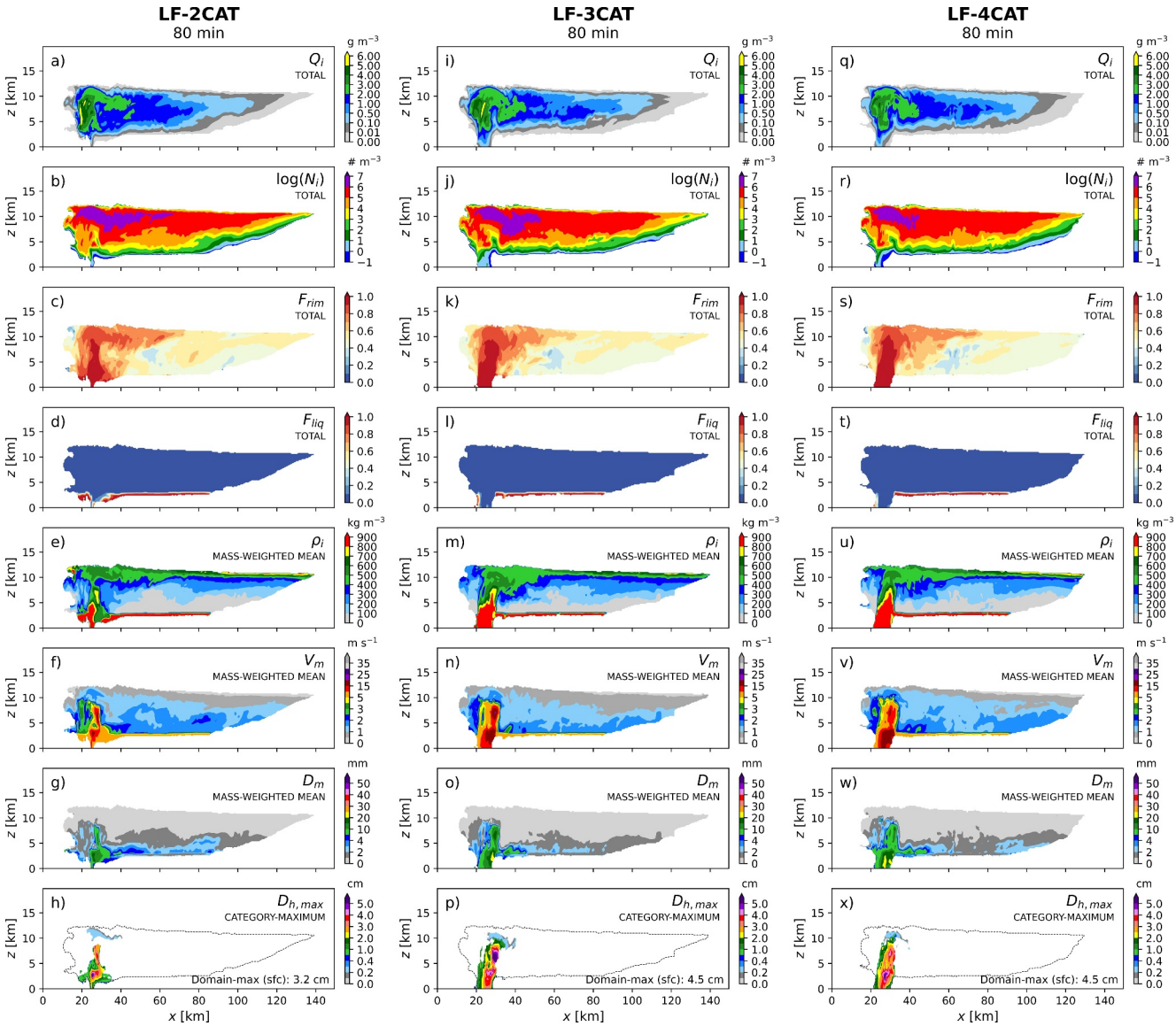


Figure 13. Bulk ice properties from LF-2CAT (left), LF-3CAT (middle), and LF-4CAT (right) (80 min). The panels in each column are similar to those in the right column of Figure 5, but are the sum (for Q_i and N_i) across all categories, the values computed from the appropriate sums (for F_{rim} and F_{liq}), mass-weighted values (for ρ_i, V_m, D_m, and), or category-maximum value ($D_{h,\text{max}}$) considering each ice category.

categories will compare better to observations compared to a run with the BASE configuration. This is because the results from any numerical simulation come from the sum of all components of the model and this invariably includes compensating errors, both in the overall atmospheric model itself and within the microphysics scheme. For melting of ice, the physics is better represented by a scheme that can model mixed-phase particles and the gradual transfer of ice to liquid within the particles followed by shedding. So, while the predicted liquid fraction results in reduced hail at the surface, which could be potentially detrimental for a given real-case simulation in terms of comparison to observations, the imacts of the specific *processes* of melting and shedding themselves are most likely improved. For the single-category configuration of P3, SIP is deactivated; this is a deliberate introduction of an "error" (the suppression of a real physical process) in order to compensate for the error of large property dilution. In the single category BASE configuration with SIP activated, the error (limitation) is the representation of all ice in the updraft with a single set of physical properties at a given point in time and space (i.e., in a grid cell), including fall speed, which has the effect of limiting hail growth to large sizes. It this case,

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there is no relevant compensating error but the error associated with dilution is mitigated with the use of multiple ice categories. Since all ice categories in multi-category configurations in P3 are subject to the identical parameterized processes and since additional categories reduce property dilution and hence improve the representation of the overall total ice, simulations should—in principle—always improve with an increased number of categories, at least with regards to the realistic modeling of ice.

The multi-category P3 configuration could be optimized such that there is a higher degree of "spread" of the properties amongst the various categories and thus a greater mitigation of property dilution. In the multi-category configuration, newly initiated ice (by primary nucleation, drop/droplet freezing, or SIP) and ice in existing categories are compared in terms of overall similarity of bulk properties. New ice is added to the category that is most similar in terms of particle properties or to a category containing no ice if one is available. At the end of the time step, ice categories that are evaluated as being similar are merged into a single category in order to free up the category for new ice. In the current version of P3, D_m is used as the proxy to evaluate overall similarity, where two populations of ice (i.e., two categories or a given category and new ice) are deemed to be similar if their difference in D_m is below a certain threshold (which is a prescribed parameter but is a decreasing function of the number of categories; see Milbrandt & Morrison, 2016). The optimal similarity condition depends on the number of ice categories used and on the type of weather being simulated, since the ultimate purpose of multiple categories is to maximize the reduction of property dilution. In order to simulate large hail optimally with multiple categories, one needs to be able to distinguish ice that represents small graupel from ice that represents large hail. In this regard, the use of D_m as a proxy for similarity is reasonable (as per the discussion above pertaining to Figure 9), though it could likely be improved. In contrast, for a large-scale baroclinic system with relatively little riming, the ability to distinguish graupel from hail is unnecessary so optimization likely means maximizing the ability to separate unrimed ice of different size ranges and/or unrimed ice from rimed ice of similar mean sizes. The proxy for overall property similarity based on D_m means that a population of high-density, heavily-rimed ice could be classified (by the current algorithm) as being similar to a population of low-density, unrimed ice with a similar mean size and hence could be merged into a single category, thereby violating the principle of minimizing property dilution. In a future study, we will explore and attempt to design an improved multi-ice-category configuration of P3, optimized for a wide range of category numbers and types of weather. We anticipate that this will use V_m as a similarity proxy and with three categories to ensure the separation of unrimed ice, graupel, and hail.

Despite potential weaknesses in the current multi-ice-category configuration, it is important to recognize that a suboptimal multi-category configuration of P3 is still a conceptual and effective improvement compared to the one-category configuration (though it could, in principle, lead to a worse solution due to compensating model errors, as discussed above), but one with less improvement than is possible. This is simply a logical deduction from the fact that all ice categories are treated with identical parameterized process and the premise that the separation of ice into generic, freely-evolving categories in P3 reduces property dilution and improves the representation of ice-phase and mixed-phase hydrometeors. There are two extremes in which a multi-category configuration would be least optimal. In the first, if the similarity condition is such that all newly-formed ice goes into the first available category such that there is never any ice in any other category then the solution will be identical to 1CAT. In the second, all newly-formed ice gets nearly evenly distributed amongst all categories resulting in virtually identical sets of properties—and thus subsequent growth rates—in each category, therefore the solution would also be very similar to 1CAT (with tiny difference arising due to changes in order-of-operations calculations, etc.). Thus, any multi-category solution in P3 that is different from the one-category solution should be more physically realistic since it the differences result from being affected by multiple (and different) ice modes and modeling their evolution independently.

Although in this study we extol the benefits of multiple ice-phase categories, this should not be interpreted as a proposal to return to traditional pre-defined category-based schemes. In P3, the categories are not in any way predefined; they all are subject to the identical parameterized process rates and can each represent any type of ice-phase or mixed-phase particle depending on the history of conditions at different locations and times. The merging of two P3 ice categories into one is not comparable to, for example, the conversion from snow to graupel in a traditional scheme. Despite the use of fixed parameters as a proxy for similarity, *merging is not conversion* since it is done between categories that are deemed to be similar, rather than being done to produce a hydrometeor type that is different, as in traditional schemes. A suboptimal P3 configuration, with a poorly chosen similarity criterion, is bound to be less harmful (i.e., result in less sensitivity) than traditional conversion processes based on

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arbitrary and unconstrained thresholds as was shown, for example, in Morrison and Milbrandt (2011) where there was a large sensitivity of supercell simulations to whether the traditional BMS favored graupel or hail.

Perhaps somewhat ironically, in addition to illustrating the benefits of multi-category P3 for the simulation of hail, the results presented in Section 4.2 also strongly support the conclusions originally made in Morrison and Milbrandt (2015) and Morrison et al. (2015) regarding the versatility of using a single generic ice-phase category in a BMS to represent a wide range of hydrometeor types and a majority of the important microphysical processes (excluding SIP). This is evident from the results which show that, except for hail in the convective core and near the surface and the number concentration at mid-levels in the convective region, the overall bulk ice properties in the 1-category, LF-off, SIP-off baseline simulation (BASE, Figure 4) are similar to those in the LF-on, SIP-on, multi-category runs (Figure 13).

6. Conclusions

This study examined the impacts of predicting the liquid fraction and multiple generic ice-phase categories in the P3 microphysics scheme in the context of cloud-resolving, hailstorm simulations for an idealized case. An important effect of predicting the liquid fraction is to improve the parameterization of melting of ice, in which a portion of the melted mass is temporarily retained in the mixed-phase particles and then gradually shed, depending on the degree of riming. This increases the rate at which ice mass is transferred to rain for heavily-rimed ice for this set of processes and ultimately reduces the amount of hail reaching the surface. In contrast, the impacts of adding ice-phase categories tend to increase the amount of hail quantities and sizes reaching the surface. This is due to reduced property dilution and the ability to allows ice to be represented in at least one category as large hail which can reside in the hail growth zone for longer times and thus grow even larger. With multiple categories the dilution of properties is reduced; multiple modes can coexist without loss to either of their sets of properties (with the exception of changes due to category interactions via aggregation) and each population of ice can continue to evolve accordingly.

For conciseness, we have limited the results presented in this paper to a single idealized case. However, for this study we also conducted the same sets of simulations and analyses (not shown) for another idealized case based on a sounding with a much different pre-storm environment, with much higher CAPE, a warmer cloud base (thereby favoring frozen drop embryos), and weaker mid-level shear; this was based on the case from Oklahoma on 1 June 2008 (also examined in Mansell et al. (2020) and Milbrandt et al. (2021)). The impacts of predicted liquid fraction and the increase of the number of ice categories were very similar to the Alberta-based case presented here in terms of the impacts of the sensitivity tests. This lends support to the generality of the conclusions regarding the systematic impacts discussed.

Since its inception, the P3 scheme has advanced in terms of its capacity to simulate hail. This includes the implementation of triple-moment ice which controls excessive size sorting, the predicted liquid faction which improves the representation of wet growth, melting, and shedding, and the use of multiple free ice categories which allows hail to grow to large sizes without dilution from smaller particles. We have recently extended the triple-moment treatment of ice to include the formal treatment of all microphysical processes, thereby improving all process rates that impact the dispersion of the size spectra; this will be reported in an upcoming study. Future improvements pertaining to hail will include optimizing the initiation and merging of ice in the multicategory configuration. On-going improvements to the representation of primary ice nucleation, other SIP mechanisms (in addition to rime splintering), and the representation of aerosols is on-going. The P3 scheme is therefore nearing the point that it can be used as a tool to conduct detailed process studies on hailstorms and other types of weather systems.

In addition to using P3 as a research tool, all of the results and conclusions on the impacts on simulating hail in this study are valid for km-scale NWP. While all of the simulations presented were done with a 200-m grid spacing to minimize interpretation challenges associated with unresolved deep convection, all of the simulations discussed were also done with a 1-km horizontal grid spacing km (not shown). Each of the sensitivities exhibited at 200 m were similar at 1 km. This is at the scale of some current real-time experimental NWP systems and is at the scale that operational systems are moving toward, even over large areas such as the CONUS. While the configuration of P3 with triple-moment ice, predicted liquid fraction, and two ice categories requires that 16 hydrometeor tracers must be advected and diffused (16 for ice, 4 for double-moment cloud and rain), which would typically be considered computationally prohibitive for operational NWP, microphysics schemes like P3 which have several

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variables per category lend themselves very well to advection techniques such as the scaled flux vector transport method (SFVT; Morrison et al., 2016). For this specific configuration of P3, 16 microphysics tracers could be advected for the computational cost of advecting four primary tracers (one per hydrometeor category) with the standard advection algorithm, plus a relatively small additional cost to compute the advective tendencies of the 12 secondary tracers. Also, due to its extensive use of pre-computed look-up tables, the P3 code itself is quite fast compared to other BMSs (see Morrison et al., 2015). We close in commenting, therefore, that the predictive detail in the LF-2CAT (or even LF-3CAT) configuration could be easily be affordable for a km-scale operational NWP system, even with current computational constraints, using an efficient advection algorithm such as SFVT or artificial-intelligence based advection.

Appendix A

Table A1
List of Symbols

Symbol	Description	Units
$B_{i,\mathrm{rim}}$	Rime ice volume mixing ratio (for a given category)	$\mathrm{m^3\ kg^{-1}}$
D	Equivalent diameter	m
$D_{h,\mathrm{max}}$	Maximum diagnosed hail diameter	m
D_m	Mass-weighted mean ice diameter	m
D_{mm}	Mean-mass diameter	m
F_{liq}	Liquid fraction (for a given ice category)	–
F_{rim}	Rime ice mass fraction (for a given category)	–
$N(D)$	Number density function (particle size distribution, PSD)	$\mathrm{\#\ m^{-1}\ kg^{-1}}$
N_0	Intercept parameter (of gamma PSD)	$\mathrm{m^{-(1+\mu)}\ kg^{-1}}$
$nCat$	Number of ice categories	–
N_i	Total ice number concentration (for a given category)	$\mathrm{\#\ m^{-3}}$
$N_{i,\mathrm{tot}}$	Total ice number mixing ratio (for a given category)	$\mathrm{\#\ kg^{-1}}$
Q_c	Mass content/concentration of cloud (droplets)	$\mathrm{kg\ m^{-3}}$
Q_i	Total ice mass concentration/content (for a given category)	$\mathrm{kg\ m^{-3}}$
$Q_{i,\mathrm{liq}}$	Liquid (on ice) mass mixing ratio (for a given category)	$\mathrm{kg\ kg^{-1}}$
$Q_{i,\mathrm{rim}}$	Rime mass mixing ratio	$\mathrm{kg\ kg^{-1}}$
$Q_{i,\mathrm{tot}}$	Total ice mass mixing ratio (for a given category)	$\mathrm{kg\ kg^{-1}}$
Q_r	Mass content/concentration of rain	$\mathrm{kg\ m^{-3}}$
V_m	Mass-weighted fall speed for ice	$\mathrm{m\ s^{-1}}$
w_{max}	Peak updraft speed	$\mathrm{m\ s^{-1}}$
Z_{DR}	Differential reflectivity (from dual-polarization radar)	dB
Z_e	Equivalent reflectivity (from model)	dBZ
$Z_{e,c}$	Equivalent reflectivity, column-maximum	dBZ
Z_H	Horizontal reflectivity (from radar)	dBZ
$Z_{i,\mathrm{tot}}$	Reflectivity factor (6th moment) mixing ratio (for a given category)	$\mathrm{m^6\ kg^{-1}}$
ρ_a	Air density	$\mathrm{kg\ m^{-3}}$
ρ_{i}	Bulk ice density	$\mathrm{kg\ m^{-3}}$
λ	Slope parameter (of gamma PSD)	$\mathrm{m^{-1}}$
μ	Shape parameter (of gamma PSD)	–

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Data Availability Statement

The code for the CM1 model is available at https://www2.mmm.ucar.edu/people/bryan/cm1. The code for the P3 scheme is available at https://doi.org/10.5281/zenodo.14659404; ongoing updates to P3 are also available at https://github.com/P3-microphysics/P3-microphysics. The model soundings and namelist files used to generate the simulations in this study are available at https://doi.org/10.5281/zenodo.14659705.

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